Roberto **Verdecchia**

ASSISTANT PROFESSOR IN SOFTWARE ENGINEERING
UNIVERSITY OF FLORENCE - SCHOOL OF ENGINEERING

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Summary

Tenure Track Assistant Professor (RTDb) at the Software Technologies Lab of the University of Florence - School of Engineering.

My research interest focuses on how empirical software-engineering methods can be used to make increasingly complex AI and digital systems more sustainable, maintainable, and architecturally responsible.

Education

Vrije Universiteit Amsterdam (VU) & Gran Sasso Science Institute (GSSI)
Amsterdam, The Netherlands & L'Aquila, Italy

DOUBLE PH.D. IN SOFTWARE ENGINEERING — *Nov. 2016 - Sept. 2021*

- Dissertation title: "Architectural Technical Debt: Identification and Management" (Assessment: *Excellent*)
- Promotors: Patricia Lago, Rocco de Nicola. Co-Promotors: Ivano Malavolta, Catia Trubiani

Vrije Universiteit Amsterdam (VU)
Amsterdam, The Netherlands

M.SC. IN COMPUTER SCIENCE: SOFTWARE ENGINEERING CURRICULUM — *2014 - 2016*

- Thesis title: "Static and Dynamic Analysis of Software Energy Efficiency in Industry" (*Cum Laude* distinction)
- Supervisors: Patricia Lago, Giuseppe Procaccianti

University of Florence - School of Engineering (UniFi)
Florence, Italy

B.ENG. IN COMPUTER ENGINEERING — *2011 - 2014*

- Thesis title: "Algorithm for Weather-based Naval Routing as an N-stage Decisional Process"
- Supervisor: Fabio Schoen

Qualifications

Italian Ministry of Education, University and Research
Italy

ITALIAN NATIONAL SCIENTIFIC QUALIFICATION (ASN) — *Oct. 2022 - Oct. 2032*

- Italian National Scientific Qualification (ASN) for Associate Professor in Information Processing Systems (ING-INF05-SC09/H1-"Sistemi di Elaborazione dell' Informazione", seconda fascia)

Work Experience

University of Florence
Florence, Italy

TENURE TRACK ASSISTANT PROFESSOR (RTDB) — *Sept. 2024 - Present*

- Assistant Professor in Software Engineering. Conducting research and teaching activities primarily focusing on the fields of Software Architecture, Technical Debt, Software Testing, and Software/AI Sustainability.

University of Florence
Florence, Italy

ASSISTANT PROFESSOR (RTDA) — *Feb. 2023 - Aug. 2024*

- Assistant Professor in Software Engineering. Conducting research and teaching activities primarily focusing on the fields of Software Architecture, Technical Debt, Software Testing, and Software/AI Sustainability.

Vrije Universiteit Amsterdam
Amsterdam, The Netherlands

RESEARCH ASSOCIATE (RTDA) — *Nov. 2019 - Feb. 2023*

- "Onderzoeker", equivalent to "Ricercatore (RTDa)" according to the table of correspondence for academic positions published by the MIUR. For more information see: https://www.miur.gov.it/news/-/asset_publisher/iN8BfUdw8tJl/content/tabella-di-corrispondenza-posizioni-accademiche.
- Responsibilities: Conduct activities following personal research vision, and support group-level research agenda. Supervise and conduct research projects, both intra- and inter-research groups. Collaborate with industrial parties (e.g., set up a research lab at ABN AMRO).

ABN AMRO
Amsterdam, The Netherlands

RESEARCH LAB COORDINATOR — *Feb. 2023 - Present*

- Coordinating, supervising, and conducting research projects in collaboration with teams of students and practitioners. Fostering research excellence, cooperation, and communication. Focusing on developing novel techniques for the assessment and optimization of the software sustainability of a portfolio comprising 600+ software products.

KPMG *Amsterdam, The Netherlands*

Graduate Research Intern *Apr. 2016 - Oct. 2016*

- Graduate research intern focusing on the analysis of a software portfolio
- Responsibilities: Architecture reconstruction. Static and dynamic analysis of code bases. Server management. Performance testing. Automation scripting. Statistical data analysis.

OMALA *Amsterdam, The Netherlands*

Software Engineering Intern *Jun. 2015 - Aug. 2015*

- Software engineering intern focusing on the analysis of a large scale heterogeneous system of systems
- Responsibilities: Requirements engineering and architecture trade-off analysis for the Airport Garden City IT infrastructure project of the Lelystad Airport business park.

Teaching Experience

Deeply committed in contributing to educational services and making a positive impact in the field of software engineering education.

Finding great satisfaction in continuously studying and innovating teaching methods, striving for evidence-based excellence in education.

Supervised award-winning theses, such as the *Dutch National Association for Software Engineering (VERSEN) MSc Best Thesis Award 2022* and the *Best Vrije Universiteit Amsterdam BSc Thesis Award 2021*.

Co-authored the serious game *DecidArch*, created to teach concepts related to software architecture, which was awarded by two distinct juries with two recognitions at the Hawaii International Conference on System Sciences (HICCS, rank A).

Regular program committee member of the research track at the International Conference on Software Engineering - Software Engineering Education and Training (ICSE-SEET) conference.

Involved in the project *Software Engineering Education and Training: Exploring Factors for Excellence*, funded from the Fondazione CR Firenze, aimed to design, optimize, and consolidate innovative educational processes in the field of software development.

University of Florence *Florence, Italy*

Lecturer *2024 - Present*

- Courses:
  - *Software Engineering and Databases*. BSc. Editions: 2025 - Current.
  - *Software Architectures and Methodologies*. MSc. Editions: 2024 - Current. Co-Instructor: Enrico Vicario.

Thesis Supervisor

- Supervisor (first or daily supervisor) of 11 BSc theses (15 ECTS each) and 16 MSc theses (30 ECTS each), including the *Best Paper Award at ECSA'24* (Sofia Migliorini, MSc 2024), the *Florentine Order of Engineers Best Thesis Award 2023* (Kevin Maggi, MSc 2023), the *Dutch National Association for Software Engineering (VERSEN) MSc Best Thesis Award 2022* (Robin van der Wiel), and the *Vrije Universiteit Amsterdam Best Thesis Award 2021* (Andrew Tan).
- Responsibilities: Identification of thesis topic, design of appropriate research methods to be used, daily supervision of theses execution, coordination of weekly theses meeting, resolution of research impediments, final theses assessment, rework and publication of theses as scientific papers.
- Theses frequently translated to scientific publications, e.g.,
  - S. Migliorini, R. Verdecchia, I. Malavolta, P. Lago, E. Vicario. *Architectural Views: The State of Practice in Open-Source Software Projects*. European Conference on Software Architecture (ECSA), 2024. 🏆 Best Paper Award.
  - K. Maggi, R. Verdecchia, L. Scommegna, E. Vicario. *CLAIM: a Lightweight Approach to Identify Microservices in Dockerized Environments*. International Conference on Evaluation and Assessment in Software Engineering (EASE), 2024.
  - N. Kozanidis, R. Verdecchia, E. Guzmàn. *Asking about Technical Debt: Characteristics and Automatic Identification of Technical Debt Questions on Stack Overflow*. International Symposium on Empirical Software Engineering and Measurement (ESEM), 2022.
  - S. Ospina, R. Verdecchia, I. Malavolta, and P. Lago. *ATDx: A tool for Providing a Data-driven Overview of Architectural Technical Debt in Software-intensive Systems*. European Conference on Software Architecture (ECSA), 2021.
  - Sophie Vos, P. Lago, R. Verdecchia, and I. Heitlager. *Architectural Tactics to Optimize Software for Energy Efficiency in the Public Cloud*. International Conference on ICT for Sustainability (ICT4S), 2022.

Vrije Universiteit Amsterdam
Amsterdam, The Netherlands

TEACHING COORDINATOR AND ASSISTANT — *2015 - Present*

- Teaching assistant and/or coordinator for a total of 18 editions of 7 BSc and MSc Computer Science courses.
  Primary responsibilities covered holding lectures, supervising research and course projects, coordinating teaching assistants, tutoring teams of students, conducting seminars and exercise sessions, designing and correcting exams/deliverables, supervising exam/deliverables correction, managing e-learning platforms, attending weekly meetings with other instructors to align course content, material, and grading criteria.
- Courses:
  - *Software Architecture* MSc. 6 ECTS. Editions: 2015, 2018, 2019.
  - *Service Oriented Design*. MSc. 6 ETCS. Editions: 2015, 2017, 2021, 2022.
  - *Literature Study and Seminar*. MSc. 6 ETCS. Editions 2018, 2019, 2020, 2021, 2022.
  - *Digitalization and Sustainability*. MSc. 6 ECTS. Editions: 2020, 2021, 2022).
  - *Software Testing* MSc. 6 ECTS. Editions: 2016, 2017.
  - *Software Modeling* BSc. 6 ECTS. Edition: 2016.
  - *Service Science*. BSc. 6 ECTS. Edition: 2015.

Lorentz Center
Leiden, The Netherlands

PHD SCHOOL MODERATOR — *2021*

- Moderator at the *International Software Architecture PhD School (iSAPS)*, Lorentz Center, Leiden, the Netherlands. 2021.

Vrije Universiteit Amsterdam
Amsterdam, The Netherlands

PHD SCHOOL LECTURER — *2020*

- Lecturer at the PhD school *Software and Sustainability: Towards an Ethical Digital Society*, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands, 2020.

Honors & Awards

2024	**Best Paper Award**, 18th European Conference on Software Architecture (ECSA) Paper: *Architectural Views: The State of Practice in Open-Source Software Projects*	*Luxembourg*
2023	**Wiley Top Cited Article**, Wiley Interdisciplinary Reviews (WIREs), Data Mining and Knowledge Discovery Paper: *A Systematic Review of Green AI*	*Online*
2022	**Best Reviewer Award**, European Conference on Software Architecture (ECSA)	*Prague, Czech Republic*
2022	**Distinguished Reviewer Award**, ACM/IEEE International Conference on Mobile Software Engineering and System (MobileSOFT)	*Pittsburgh, U.S.A.*
2021	**Best Paper Award**, Journal of Systems And Software (JSS) Paper: *Building and evaluating a theory of architectural technical debt in software-intensive systems*	*Online*
2021	**Facebook Research Award**, Facebook research. Grant: 94.500$ USD Project: *Testing non-testable programs using association rules*	*London, United Kingdom*
2021	**Best Research Presentation Award**, International Conference on Technical Debt (TechDebt) Paper: *Characterizing Technical Debt and Antipatterns in AI-Based Systems: A Systematic Mapping Study*	*Madrid, Spain*
2019	**ACM-SIGSOFT Distinguished Paper Award**, IEEE-ACM International Conference on Software Engineering (ICSE) Paper: *Scalable Approaches for Test Suite Reduction*	*Montreal, Canada*
2019	**Facebook Testing and Verification Award**, Facebook Testing and Verification Symposium 2019 Grant: 50.000$ USD Project: *Static Prediction of Test Flakiness*	*London, United Kingdom*
2019	**Best Paper Award**, Hawaii International Conference on System Sciences (HICSS) Paper: *DecidArch: Playing Cards as Software Architects*	*Hawaii, U.S.A*
2019	**ISSIP-IBM-CBA Student Paper Award for Best Industry Studies Paper**, Hawaii International Conference on System Sciences (HICSS)	*Hawaii, U.S.A*
2018	**Best Early Career Researcher Award**, IEEE International Conference on Software Architecture (ICSA)	*Seattle, U.S.A*
2018	**Runner-up Best Paper Award**, International Conference on ICT for Sustainability (ICT4S)	*Toronto, Canada*
2018	**Bronze Medal - ACM Student Research Competition**, 5th International Conference on Mobile Software Engineering and System (MobileSoft)	*Seattle, U.S.A*

Funded Projects

ABN AMRO

GRANT: 250.000€

- Role: Responsible of research execution and supervision.
- Topic: The project aims at devising, via a combination of in-vivo and in-vitro experimentation, innovative methods to measure, track, document, and improve the sustainability of industrial software-intensive systems.

Facebook Research (now Meta Research)

GRANT: 94.500$

- Role: Co-Principal Investigator (Co-PI). Co-PIs: Breno Miranda (UFPE, Brazil), Antonia Bertolino (ISTI-CNR, Italy), Emilio Cruciani (PLUS, Salzburg).
- Project: The project aims at developing ARMED, an approach leveraging efficient association rules mining algorithms to identify plausible and suspicious behaviours in large-scale software-intensive systems when oracles are inherently unknowable.

Foundation Cassa di Risparmio di Firenze

GRANT: 50.000€

- Role: Co-Principal Investigator (Co-PI). Co-PI: Enrico Vicario (University of Florence).
- The project aims to design, optimize, and consolidate innovative educational processes in the field of software development. By involving heterogeneous pedagogical partners present in the Florentine province, e.g., the school 42 by Luiss, the project employs rigorous empirical methods to provide scientific evidence and advancements of software engineering education.

Facebook Research (now Meta Research)

GRANT: 47.250$

- Role: Co-Principal Investigator (Co-PI). Co-PIs: Breno Miranda (UFPE, Brazil), Antonia Bertolino (ISTI-CNR, Italy), Emilio Cruciani (PLUS, Salzburg).
- Project: The project aims at developing a fully automated approach to swiftly identify test flakiness via well-known static analysis techniques, to immediately return potentially flaky tests to their creators before the tests are released in a repository.

Netherlands Enterprise Agency

GRANT: 28.586€

- Role: Co-Principal Investigator (Co-PI). Co-PI: Patricia Lago (Vrije Universiteit Amsterdam).
- Topic: The aim of the project consisted in developing a landscape of solutions for the sustainable evolution of present and future digital infrastructures.

Amsterdam Sustainability Institute

GRANT: 10.000$

- Role: Co-Principal Investigator (Co-PI). Co-PIs: Henri L.F. de Groot (Vrije Universitet Amsterdam), Patricia Lago (Vrije Universitet Amsterdam).
- The project aims at uncovering tactics for sustainable clouds, by combining technical software engineering solutions for distributed clouds (from computer science) and novel business- and behavioural models for the built environment (from environmental economics).

Spanish Ministry of Science and Innovation

GRANT: 277.035€

- Role: Scientific Research Collaborator. PIs: Xavier Franch Gutieŕrez (Polytechnic University of Catalonia), Silverio Martińez-Fernańdez (Polytechnic University of Catalonia).
- Project: The project aims at designing and evaluating architecture-centric approaches to implement Green AI systems, i.e., energy efficient AI-based systems that keep other non-functional requirements (e.g., accuracy and privacy) at a satisfactory level.

Public Speaking

Conducted more than 30 invited talks and presentations at various venues. Activities consisted of presenting complex research concepts and results to audiences of different nature. Presentation targeted either specialized academic / industrial attendees, or the generic public. Selected presentation venues: Google Journal Club, International Conference on Software Engineering, KPMG Global Headquarters, International Conference on Software Architecture, University of British Columbia, University of Zurich.

Selected Invited Talks

- "Is there a road towards Greener AI?", UC Berkeley School of Law. Panel. Berkeley, U.S.A. 2025.
- "Green AI: Quando le Soluzioni Non Bastano" ("Green AI: When Solutions are Not Enough"), University of Padova. Padova, Italy. 2025.
- "Why are we not making AI Green?", Polytechnic University of Milan. Panel. Milan, Italy. 2025.
- "Automated Software Testing for Sustainability and Resource Efficiency", International Conference on Automation of Software Test (AST). Panel. Ottawa, Canada. 2025.
- "Technical Debt Seeds", Dagstuhl Perspectives Workshop 24452 - Reframing Technical Debt. Wadern, Germany. 2024.
- "Discovering the Undercurrents: A Glimpse at Technical Debt in Self-Adaptive Systems", CASA Workshop. Keynote. Luxembourg. 2024.
- "Empirical Research for Greener Software: Where Two Worlds Meet", Centro de Informática, Federal University of Pernambuco. Recife, Brazil. 2024.
- "Green Software: Unleashing The Untapped Potential", Global Education for a Sustainable Future Summit (Ge4). Trento, Italy. 2024.
- "Behind the Curtains of Software Engineering", School 42. Florence, Italy. 2024.
- "Personal Research Agenda", Scuola Superiore Sant'Anna. Pisa, Italy. 2023.
- "Tackling Architectural Technical Debt", Universität Zurich. Virtual. 2021.
- "Lower Energy Acceleration Programme (LEAP) Technology Roadmap", Groene Peper. Virtual. 2021.
- "Building and evaluating a theory of architectural technical debt in software-intensive systems", JSS Happy Hour. Virtual. 2021.
- "What to consider when talking about ATD?", ICSA. Virtual. 2021.
- "Scalable Approaches for Test Suite Reduction", Google Journal Club. Virtual. 2019.
- "Code Analysis Techniques for Architectural Technical Debt Identification", Software Improvement Group. Amsterdam, The Netherlands. 2019.
- "Building a Grounded Theory on Architectural Technical Debt", TaskTop Technologies. Vancouver, Canada. 2019.
- "Architectural Technical Debt: Personal Research Agenda", University of British Columbia. Vancouver, Canada. 2019.
- "FAST Approaches to Scalable Similarity-based Test Case Prioritization", Gran Sasso Science Institute. L'Aquila, Italy. 2018.
- "Static and Dynamic Analysis of Software Energy Efficiency", KPMG Global Headquarters. Amstelveen, The Netherlands. 2017.

Selected Media Appearances

- Bloomberg: "Big Tech's Climate Goals At Risk From Massive AI Energy Demands". 2024.
- New Books Network: discussion on architectural technical debt. Podcast. 2024.
- Physics Today: "Will AI's growth create an explosion of energy consumption?". 2024.
- National Museum of Science and Technology Leonardo da Vinci: "Environmental impact of Artificial Intelligence: what are the possible solutions?". 2024.
- New York Times: "A.I. Could Soon Need as Much Electricity as an Entire Country". 2023.
- The Wall Street Journal: "AI Is Ravenous for Energy. Can It Be Satisfied?". 2023.
- IEEE Spectrum: "Generative AI's Energy Problem Today is Foundational". 2023.
- New Scientist: "Artificial intelligence training is powered mostly by fossil fuels". 2023.
- 6Gworld: "Green AI - An Energy-Saving Paradise Lost (So Far)". 2023.
- Computable: "Geld nodig voor duurzame, digitale delta". 2021.
- La Repubblica and Open: coverage of the Facebook Testing and Verification Research Award. 2019.

Services

Organizing Committee	ICPE 2026 (*General Co-Chair*), ICSA 2026 (*Journal-First Co-Chair*), SANER 2026 (*Tool Demo Co-Chair*), TechDebt 2025 (*Program Co-Chair*), QUATIC 2025 (*GSC Thematic Track Co-Chair*), TechDebt 2024 (*Publications Chair*), QUATIC 2024 (*QUAS Thematic Track Co-Chair*), ICSA 2024 (*Publications Co-Chair*), ISSRE 2023 (*Local Co-Organizer*), QuaLAS 2023 (*Co-Organizer*), GREENS 2023 (*Web and Proceedings Chair*), MSR4SA 2022 (*Co-Organizer*), ECSA 2022 (*Proceedings Co-Chair*), BoKSS 2021 (*Virtualization Chair*).
Steering Committee	ICSA (2025 - Current)
Editorial Board	Scientific Reports (Nature)
Program Committee member	TechDebt (2021-2026), ICSE-SEET / JSEET and Artifact Evaluation (2021, 2023-2025), ICSA (2024-2026), WSESE 2026, SANER (2024), ESEM (2023-2025), FSE-SEET 2025, ECSA (2021-2022, 2025), MSR (2022), ICSME (2022), MobileSoft (2022, 2023), SAML (2022, 2024, 2026), TD4ViS (2022), ICSOFT (2021-2022), SOFTENG (2020), GREEN (2019), GREENS (2024-2026), MSR4SA (2021), SSCOPE (2023)
Journal Reviewer	JSS (2019-), TOSEM (2021-), TSE (2020-), EMSE (2020-), IST (2020-), CSR (-), SQJ (-), SUSCOM (2019-), JCS (-), COLA (2019-), IJES (2021-), JSERD (2021-)
Magazine Reviewer	IEEE Software (2021-)
Book Reviewer	Springer Nature
Academic Positions of Trust	Faculty Level (School of Engineering), Council Member (2025 - Current) Department Level (Department of Information Engineering), Inner Advisory Council Member (2025 - Current)
Italian Committees	Order of Engineers Italian State Examination, Expert Committee Member (2024)
External Conference Reviewer	ICSE (2017-2022), EASE (2022), SESoS (2022), ESEC/FSE (2017, 2021), ICSA (2018-2020), ECSA (2018-2020), MOBILESoft (2019), ICT4S (2018-2020, 2022), CaiSE (2020), ICSE-SEIS (2017)
Volunteer	ICSE (2018, 2019), ICPE (2017)

Visitings

Software and Sustainability Group (S2), Vrije Universiteit Amsterdam — *Amsterdam, The Netherlands*

VISITING RESEARCH SCHOLAR — *September 2025*

- Host: Patricia Lago

Centro de Informática, Federal University of Pernambuco — *Recife, Brazil*

VISITING RESEARCH SCHOLAR — *July 2024*

- Host: Breno Miranda

Electrical and Computer Engineering Department (University of British Columbia) — *Vancouver, Canada*

VISITING RESEARCH SCHOLAR — *April 2019 - June 2019*

- Topic: Approaches for architectural technical debt identification and management
- Host: Philippe Kruchten

SEDC Lab, ISTI-CNR (Italian National Research Council) — *Pisa, Italy*

VISITING RESEARCH SCHOLAR — *July 2017, July 2018, July 2019*

- Topic: Fast static approaches for test prioritization, selection, and flakiness prediction
- Host: Antonia Bertolino

PhD Schools

International Software Architecture PhD School (iSAPS) *Leiden, The Netherlands*

PhD School Attendee *June 2017, June 2018, June 2019*

- Topic: PhD school series on Software Architecture providing researchers and practicing architects the opportunity to learn from the leaders in the field the most recent methods, tools, and theories produced by top software architecture research and industrial groups around the world.

In-Vivo Analytics for Big Software Quality *Leiden, The Netherlands*

PhD School Attendee *September 2018*

- Topic: PhD school aimed to address how Big Data analysis techniques on runtime data could be used to address software quality in DevOps. Discussion on core challenges to develop cooperations in the related areas, e.g., log data analysis, requirements engineering, automated testing, program repair, and distributed systems.

ICT for Sustainability (ICT4S) *Leiden, The Netherlands*

PhD School Attendee *May 2017, May 2020*

- Topic: PhD school series on ICT for Sustainability (ICT4S), focusing both on how to make ICT greener, and how to use ICT to develop sustainable solutions in diverse areas. The goal of the school is to build a bridge between related communities, kickstart new scientific collaborations and nurture the next generation of community members through presentations by leading academics and collaborative paper writing.

ICT with Industry *Leiden, The Netherlands*

PhD School Attendee *November 2017, November 2019*

- Topic: PhD school bringing together junior scientists and professionals from industry and governments to collaborate on industrial hands-on projects. The workshop focused on case studies, which were subject to an intense week of analyzing, discussing, and modeling solutions.

Open Science, Study Replicability, and Open Source Software

Strong supporter of open science, study replicability, and open source software. All co-authored studies are accompanied by a replication package, containing the entirety the material necessary to replicate studies, including all the data and source code used, analysis traces, and intermediate and final results. The only exceptions are due to non-disclosure agreements with industrial partners, and/or human ethics guidelines governing studies (e.g., to anonymize interviews).

Open Science, Study Replicability, and Open Source Software (cont.)

Whenever given the opportunity, artifact evaluations are requested (e.g., the study "E. Cruciani, B. Miranda, R. Verdecchia, A. Bertolino. *Scalable Approaches for Test Suite Reduction*. International Conference on Software Engineering (ICSE), 2019" was awarded with the "Artifact available" and "Artifact evaluated-reusable" ACM badges).

Tools and source code related to publications are released as open-source software under permissive free software licenses. For example, the tool ATDx is released as open-source software under MIT license: `https://github.com/robertoverdecchia/ATDx`

A complete overview of replication packages and tools developed can be found at:

- Personal GitHub page: `https://github.com/robertoverdecchia`
- Software and Sustainability Group (VU) GitHub Page: `https://github.com/S2-group`
- Software Technology Lab GitHub page (recently established): `https://github.com/STLab-UniFI`

Languages

Italian	Native
German	Native
English	Full professional proficiency

Publications

- J. Bogner, R. Verdecchia. *The State of Peer Review in Empirical Software Engineering: A Community Survey on Review Load, Quality, and GenAI Use*. ACM SIGSOFT Software Engineering Notes, 2026.
- M. Becattini, N. Caselli, M. Minin, R. Verdecchia, E. Vicario. *CAPRA: Scaling Feedback on Software Architecture Deliverables with a Multi-Agent LLM System*. CSEE&T, 2026.
- B. Heeren, F. Dalpiaz, M. Seraj, R. Verdecchia, V. Zaytsev. *A Systematic Analysis of Higher Education on Software Engineering in the Netherlands*. Journal of Systems and Software, 2026 (forthcoming).
- R. Verdecchia, E. Sarri, E. Vicario. *"That Developer Left the Project!": An Introduction and Case Study of Turnover Technical Debt*. TechDebt, 2026.
- N. Scommegna, L. Bindini, K. Noor Ali, R. Verdecchia, S. Marinai. *Evaluating Software Architecture Diagram Connectivity with Vision Large Language Models*. Symposium on Document Engineering, 2026.
- R. Verdecchia, J. Bogner. *The Human Need for Storytelling: Reflections on Qualitative Software Engineering Research With a Focus Group of Experts*. ACM SIGSOFT Software Engineering Notes, 2026.
- R. Verdecchia, E. Cruciani, A. Bertolino, B. Miranda. *Energy-Aware Software Testing*. ICSE, 2025.
- M. Alswaitti, R. Verdecchia, G. Danoy, P. Bouvry, J.E. Pecero. *Training Green AI Models Using Elite Samples*. IEEE Transactions on Sustainable Computing, 2025.
- J. Bogner, R. Verdecchia. *ACM SIGSOFT SEN Empirical Software Engineering: Introducing Our New Regular Column*. ACM SIGSOFT Software Engineering Notes, 2025.
- E. Cruciani, R. Verdecchia. *Choosing to Be Green: Advancing Green AI via Dynamic Model Selection*. Green-Aware AI, 2025.
- G. Bertazzini, C. Albisani, D. Baracchi, D. Shullani, R. Verdecchia. *The Hidden Cost of an Image: Quantifying the Energy Consumption of AI Image Generation*. arXiv preprint, 2025.
- R. Verdecchia, J. Bogner. *Notes On Writing Effective Empirical Software Engineering Papers: An Opinionated Primer*. ACM SIGSOFT Software Engineering Notes, 2025.
- M. Becattini, R. Verdecchia, E. Vicario. *SALLMA: A Software Architecture for LLM-Based Multi-Agent Systems*. SATrends, 2025.
- K. Borowa, K. Ratkowski, R. Verdecchia. *The Technical Debt Gamble: A Case Study on Technical Debt in a Large-Scale Industrial Microservice Architecture*. Journal of Systems and Software, 2025.
- M. Becattini, R. Verdecchia, E. Vicario. *An Accountability-Based Architectural Tactic for Agent Cooperation in LLM-Based Multi-Agent Systems*. IEEE BigData, 2025.
- F. Ponce, R. Verdecchia, B. Miranda, J. Soldani. *Microservices testing: A systematic literature review*. Information and Software Technology, 2025.
- L. Scommegna, B. Picano, R. Verdecchia, E. Vicario. *OREO: A Tool-Supported Approach for Offline Run-time Monitoring and Fault-Error-Failure Chain Localization*. Journal of Systems and Software, 2025.
- J.A. Ebert, Z. Codabux, R. Verdecchia. *Racing Against the Clock: Exploring the Impact of Scheduled Deadlines on Technical Debt*. EASE, 2025.
- N. Marini, L. Pampaloni, F. Di Martino, R. Verdecchia, E. Vicario. *Green AI: Which Programming Language Consumes the Most?*. GREENS, 2025.
- L. Cruz *et al.* *Greening AI-enabled Systems with Software Engineering: A Research Agenda for Environmentally Sustainable AI Practices*. ACM SIGSOFT Software Engineering Notes, 2025.
- A. Domingo Reguero, S. Martínez-Fernández, R. Verdecchia. *Energy-efficient neural network training through runtime layer freezing, model quantization, and early stopping*. Computer Standards & Interfaces, 2025.
- K. Maggi, R. Verdecchia, L. Scommegna, E. Vicario. *Evolution of Code Technical Debt in Microservices Architectures*. Journal of Systems and Software, 2025.
- Z. Codabux, F. Fard, R. Verdecchia, F. Palomba, D. Di Nucci, G. Recupito. *Teaching Mining Software Repositories*. In *Teaching Empirical Research Methods in Software Engineering*, Springer, 2025.
- P. Lago, P. Runeson, Q. Song, R. Verdecchia. *Threats to Validity in Software Engineering–hypocritical paper section or essential analysis?*. ESEM, 2024.
- S. Migliorini, R. Verdecchia, I. Malavolta, P. Lago, E. Vicario. *Architectural Views: The State of Practice in Open-Source Software Projects*. ECSA, 2024. Best Paper Award.
- T. Trinci, S. Magistri, R. Verdecchia, A.D. Bagdanov. *How green is continual learning, really? Analyzing the energy consumption in continual training of vision foundation models*. GreenFOMO, 2024.
- N. Pollini, K. Maggi, R. Verdecchia, E. Vicario. *Learning Programming without Teachers: An Ongoing Ethnographic Study at 42*. LEARNER, 2024.
- K. Maggi, R. Verdecchia, L. Scommegna, E. Vicario. *CLAIM: a Lightweight Approach to Identify Microservices in Dockerized Environments*. EASE, 2024.
- R. Verdecchia, K. Maggi, L. Scommegna, E. Vicario. *Technical Debt in Microservices: A Mixed-Method Case Study*. ECSA Post-Proceedings, 2024.
- L. Scommegna, R. Verdecchia, E. Vicario. *Unveiling Faulty User Sequences: A Model-based Approach to Test Three-Tier Software Architectures*. Journal of Systems and Software, 2024.
- M. Funke, P. Lago, R. Verdecchia, R. Donker. *A Process for Monitoring the Impact of Architecture Principles on Sustainability: An Industrial Case Study*. Software (MDPI), 2024.
- R. Verdecchia, L. Scommegna, B. Picano, M. Becattini, E. Vicario. *Network Digital Twins: A Systematic Review*. IEEE Access, 2024.
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